Can you read your resources, lectures pdf files and transcripts and tell me if my plan for exam is good enough and what should i do and how should i study?
if you wanna go into details, for example this is the task i'm doing today. how do i do it? say in detail and i'll add it to my task and do it.
my problem is that im not sure if my learning material is limited to the lecture slides and practices.

Thought for 2m 14s

Answer

Yes, your **overall exam plan is good enough**, but only under one condition: you must treat **tutorial problems + solutions as the centre of MAT700 revision**, not the lecture slides alone.

For MAT700, the folder I can see contains **Lecture 1–7 PDFs, Lecture 1 supplement, Tutorial 1–3 PDFs, tutorial solutions, MNIST demo, module description, and class transcripts**. I **can't find MAT700 past papers in the provided MAT700 materials yet**, so your practical exam training has to come mainly from **Tutorial 1–3, their solutions, and timed re-drills**.

Your plan already puts MAT700 into a rapid foundation phase, then a crunch week before the 11 May exam, which is sensible. Your Todoist task for today is specifically: **"MM L1 read + Tutorial 1 P1–P3"**, 19:00–20:30, covering Lecture 1 plus the birthday problem and IDF/TF-IDF computations. Todoist Study Plan - Strict Str…

The main fix: **do not spend the whole 90 minutes beautifully reading all 44 slides.** Lecture 1 is partly conceptual and partly repeated/animated. You need enough understanding to do the tutorial problems.

How to do today's task: MM L1 + Tutorial 1 P1–P3

0–5 min — Set up

Open:

- `Lecture1.pdf`

- `Tutorial1-beamer-version.pdf`
- `MA3700SolutionsTutorial1.pdf`

Create one notebook page called:

MM L1 + Tutorial 1 P1–P3

Use this structure:

```
1. Key ideas from Lecture 1
2. P1 Birthday problem
3. P2 IDF
4. P3 TF-IDF
5. Mistakes / things to redo
```

5–45 min — Read Lecture 1 actively

Do **not** copy the slides. For each section, write only:

```
Definition:
Why it matters:
Possible exam trigger:
```

Focus on these parts:

A. What the module is about
Lecture 1 says the module introduces mathematical models and methods for data mining, including Bayesian inference, optimisation, collaborative filtering, clustering, pattern analysis, kernel methods, and learning theory. It also says the module is assessed by **100% written examination**.

Write:

```
MAT700 is about mathematical models for extracting patterns/knowledge from data.
Assessment = 100% written exam.
```

B. What data mining means
Lecture 1 defines data mining as extracting patterns from large data sets by combining statistics, AI, and database management.

Write:

```
Data mining = extracting patterns from large datasets.
```

```
Output = models or patterns.
```

C. Five main application types

Lecture 1 lists:

```
classification
numerical prediction
association
clustering
ranking
```

For today, just recognise them. Do not deep study all of them yet.

D. Bonferroni / birthday idea

This matters because Tutorial 1 P1 is basically an expected-count problem. The key idea is:

```
When searching huge data, random coincidences can happen.
So we compute the expected number of coincidences under randomness.
```

E. Labelled vs unlabelled data

Write:

```
Labelled data → supervised learning.
Categorical label → classification.
Numerical label → regression.
Unlabelled data → unsupervised learning.
```

Stop Lecture 1 after 40 minutes even if it feels incomplete. The tutorial matters more.

45–60 min — Tutorial 1 P1: birthday problem

The problem asks for the expected number of triples of students with the same birthday in a class of 120.

The hint says: **K objects in N boxes**, where:

```
K = 120 students
N = 365 days
m = 3 students in the same box/day
```

Use the formula:

```
Expected number of m-tuples in the same box =
C(K, m) / N^(m-1)
```

So:

```
C(120, 3) / 365^2
```

Now compute:

```
C(120, 3) = 120 × 119 × 118 / 6 = 280840
365^2 = 133225
280840 / 133225 ≈ 2.108
```

Meaning:

```
In a random class of 120 students, we expect about 2.108 triples of students sharir
```

The real understanding:

```
We are not asking whether one triple exists.
We are asking how many triples we expect on average.
```

60–73 min — Tutorial 1 P2: IDF

Formula:

```
IDF_i = log₂(N / n_i)
```

Where:

```
N = total number of documents
n_i = number of documents containing the word
```

P2(a)

```
N = 10,000,000
n_i = 40
IDF = log₂(10,000,000 / 40)
    = log₂(250,000)
    ≈ 17.93
```

Nearest integer:

```
18
```

P2(b)

```
N = 10,000,000
n_i = 10,000
IDF = log₂(10,000,000 / 10,000)
    = log₂(1000)
    ≈ 9.97
```

Nearest integer:

```
10
```

Interpretation:

```
Rare word → higher IDF.
Common word → lower IDF.
```

73–85 min — Tutorial 1 P3: TF-IDF

P3 gives:

```
N = 10,000,000
n_i = 320
max word occurrence in document D = 15
```

First compute IDF:

```
IDF = log₂(10,000,000 / 320)
    = log₂(31,250)
    ≈ 14.93
```

Formula for term frequency:

```
TF_ij = f_ij / max_k(f_kj)
```

P3(a): word appears once

```
TF = 1 / 15
TF-IDF = (1/15) × 14.93
```

```
       ≈ 0.995
       ≈ 1.0
```

P3(b): word appears five times

```
TF = 5 / 15 = 1/3
TF-IDF = (1/3) × 14.93
       ≈ 4.98
       ≈ 5.0
```

Interpretation:

```
Same word, same IDF.
Appearing 5 times gives 5 times the TF score compared with appearing once.
```

85–90 min — Error log

Write this at the bottom:

```
Things I must remember:
- Birthday problem uses expected count, not probability of at least one event.
- IDF uses log base 2.
- TF uses occurrence of the word divided by max occurrence of any word in that docu
- TF-IDF = TF × IDF.
- Round only when the question asks for approximate / nearest integer.
```

Copy-paste version for your Todoist task description

```
Goal: Finish MM Lecture 1 actively + Tutorial 1 P1–P3.

Files:
- Lecture1.pdf
- Tutorial1-beamer-version.pdf
- MA3700SolutionsTutorial1.pdf

Method:
1. Read Lecture 1 for 40 min max. Do not copy slides. Write only:
   - definition
   - why it matters
   - possible exam trigger

Focus on:
- data mining definition
- 100% written exam
```

```
- five application types: classification, numerical prediction, association, cluste
- Bonferroni / birthday idea
- labelled vs unlabelled data
- classification vs regression
- unsupervised learning and outliers

2. Tutorial 1 P1:
   - Use expected number formula for K objects in N boxes.
   - Compute C(120,3)/365^2 ≈ 2.108.
   - Write one sentence explaining why this is expected count, not probability.

3. Tutorial 1 P2:
   - Use IDF = log2(N/n_i).
   - For 40 docs: log2(250000) ≈ 18.
   - For 10000 docs: log2(1000) ≈ 10.

4. Tutorial 1 P3:
   - First compute IDF = log2(10000000/320) ≈ 14.93.
   - If word appears once: TF = 1/15, TF-IDF ≈ 1.0.
   - If word appears five times: TF = 5/15, TF-IDF ≈ 5.0.

5. Check solutions.
6. Add mistakes to error log.
```

Is your learning material limited to slides and practices?

For **exam preparation**, practically yes:

```
Core = lecture PDFs + tutorial sheets + tutorial solutions.
Support = class transcripts/recordings.
Supplementary = MNIST notebook and images, mainly for understanding/demo.
```

The module description says teaching may include lectures/classes plus electronic resources like videos, exercise sheets, lecture notes, and quizzes, and expects at least 50 hours of self-guided study. So you should check Learning Central for any quizzes or extra exercise sheets, but in the materials I currently see, the main examinable revision base is the lectures and tutorials.

Use transcripts like this:

```
When a slide feels unclear → read the matching transcript section.
When you understand the slide → do not spend time on the transcript.
```

The transcripts are good for explanations, but they are too long and messy to be your main study material.

Common mistakes

- Reading slides passively and feeling “done” without solving tutorial problems.
- Spending too long on Lecture 1. It is important, but not as calculation-heavy as later topics.
- Forgetting that Tutorial 1 P1 is an **expected value** problem.
- Using natural log or log base 10 instead of `log₂` in IDF.
- Mixing up `n_i` and `N`: `N` is all documents, `n_i` is documents containing the word.
- Not making an error log. Your error log is what becomes your final revision sheet.

Quick check

1. Why is the birthday answer about **expected number of triples**, not “probability of a triple existing”?
2. If a word appears in fewer documents, does its IDF go up or down?
3. In TF-IDF, what does the denominator `max_k f_kj` mean?

Sources

- `Lecture1.pdf`, p.5–6, p.9, p.24–28, p.32–37
- `Tutorial1-beamer-version.pdf`, p.2–4
- `MAT700 Mathematical Methods for Data Mining module description.docx`, no page numbering in file
- `Todoist Study Plan - Strict Structure.md`, MAT700 Phase 1 task list

Todoist Study Plan - Strict Str...

↳ How to do today’s task: MM L1 + Tutorial 1 P1–P3 ---------- ### 0–5 min — Set up Open: * `Lecture1.pdf` * `Tutorial1-beamer-version.pdf` * `MA3700SolutionsTutorial1.pdf` Create one notebook page called: **MM L1 + Tutorial 1 P1–..

so i added your description to the task description but i need a breakdown into subtasks the way you described at first. (0–5 min — Set up, 5–45 min — Read Lecture 1 actively, ..., 85–90 min — Error log)
give for each of them title, labels, description, time and duration, priority and every other thing neccessary

Thought for 27s ›

Yes — make them **subtasks under the existing parent task**, not separate main tasks. Since your parent task is already **P2 @core @lecture**, the subtasks can stay simple and structured.

Assuming the parent block is **19:00–20:30**, add these:

Subtasks to add in Todoist

1. Set up MM L1 workspace

Title:
`Set up MM L1 workspace`

Time: 19:00
Duration: 5 min
Priority: P4
Labels: `core`
Reminder: none
Location: same as parent / leave blank

Description:

```
Open:
- Lecture1.pdf
- Tutorial1-beamer-version.pdf
- MA3700SolutionsTutorial1.pdf

Create one notebook page:
MM L1 + Tutorial 1 P1–P3
```

```
Use sections:
1. Key ideas from Lecture 1
2. P1 Birthday problem
3. P2 IDF
4. P3 TF-IDF
5. Mistakes / things to redo
```

---

2. Read Lecture 1 actively

Title:

`Read Lecture 1 actively`

Time: 19:05

Duration: 40 min

Priority: P2

Labels: `core`, `lecture`

Reminder: none

Location: same as parent / leave blank

Description:

```
Read Lecture 1 for understanding, not copying.

For each important section, write only:
- Definition
- Why it matters
- Possible exam trigger

Focus on:
- data mining definition
- 100% written exam
- five application types:
  classification, numerical prediction, association, clustering, ranking
- Bonferroni / birthday idea
- labelled vs unlabelled data
- classification vs regression
- unsupervised learning and outliers

Important rule:
Stop after 40 min even if the lecture feels unfinished. Tutorial practice matters m
```

---

3. Do Tutorial 1 P1: birthday problem

Title:

`Do Tutorial 1 P1 birthday problem`

Time: 19:45

Duration: 15 min

Priority: P2

Labels: `core`

Optional label if you created it: `exercise`

Reminder: none

Location: same as parent / leave blank

Description:

```
Goal:
Understand expected number of triples with the same birthday.

Use:
K = 120 students
N = 365 days
m = 3 students

Formula:
Expected number = C(K,m) / N^(m-1)

Compute:
C(120,3) / 365^2 ≈ 2.108

Write one sentence:
This is an expected count, not the probability that at least one triple exists.
```

---

4. Do Tutorial 1 P2: IDF computation

Title:

`Do Tutorial 1 P2 IDF computation`

Time: 20:00

Duration: 13 min

Priority: P2

Labels: `core`

Optional label: `exercise`

Reminder: none

Location: same as parent / leave blank

Description:

```
Formula:
IDF_i = log2(N / n_i)

Where:
N = total number of documents
n_i = number of documents containing the word

P2(a):
N = 10,000,000
n_i = 40
IDF = log2(250,000) ≈ 17.93 ≈ 18

P2(b):
N = 10,000,000
n_i = 10,000
IDF = log2(1000) ≈ 9.97 ≈ 10

Write interpretation:
Rare word → higher IDF.
Common word → lower IDF.
```

---

5. Do Tutorial 1 P3: TF-IDF computation

Title:

`Do Tutorial 1 P3 TF-IDF computation`

Time: 20:13

Duration: 12 min

Priority: P2

Labels: `core`

Optional label: `exercise`

Reminder: none

Location: same as parent / leave blank

Description:

```
Given:
N = 10,000,000
n_i = 320
max word occurrence in document D = 15

First compute:
IDF = log2(10,000,000 / 320)
```

```
    = log2(31,250)
    ≈ 14.93

Formula:
TF_ij = f_ij / max_k(f_kj)

P3(a): word appears once
TF = 1/15
TF-IDF = (1/15) × 14.93 ≈ 1.0

P3(b): word appears five times
TF = 5/15 = 1/3
TF-IDF = (1/3) × 14.93 ≈ 5.0

Write interpretation:
Same word, same IDF.
Appearing 5 times gives 5 times the TF score compared with appearing once.
```

---

6. Check solutions and add error log

Title:

`Check solutions and add error log`

Time: 20:25

Duration: 5 min

Priority: P4

Labels: `core`, `review`

Reminder: none

Location: same as parent / leave blank

Description:

```
Open MA3700SolutionsTutorial1.pdf.

Check:
- P1 birthday problem
- P2 IDF
- P3 TF-IDF

Add mistakes to notebook under:
Mistakes / things to redo

Write these reminders:
- Birthday problem uses expected count, not probability of at least one event.
- IDF uses log base 2.
- TF uses occurrence of the word divided by max occurrence of any word in that docu
```

```
- TF-IDF = TF × IDF.
- Round only when the question asks for approximate / nearest integer.
```

Clean Todoist version

```
Parent task:
MM L1 read + Tutorial 1 P1–P3
19:00–20:30 | P2 | @core @lecture

Subtasks:
1. Set up MM L1 workspace
   19:00 | 5 min | P4 | @core

2. Read Lecture 1 actively
   19:05 | 40 min | P2 | @core @lecture

3. Do Tutorial 1 P1 birthday problem
   19:45 | 15 min | P2 | @core

4. Do Tutorial 1 P2 IDF computation
   20:00 | 13 min | P2 | @core

5. Do Tutorial 1 P3 TF-IDF computation
   20:13 | 12 min | P2 | @core

6. Check solutions and add error log
   20:25 | 5 min | P4 | @core @review
```

Small Todoist tip: don't add reminders to these subtasks. Your plan's rule says reminders are mainly for P1 events, while P2/P3/P4 study tasks should just appear in Today view. Also keep subtask titles without emojis, because your Todoist structure explicitly says subtasks should stay uncluttered. Todoist Study Plan - Strict Str…

Sources:

- Todoist Study Plan - Strict Structure.md, MAT700 Phase 1 task and Todoist label/priority/reminder rules Todoist Study Plan - Strict Str… Todoist Study Plan - Strict Str…
- Mathematical Methods for Data Mining and Time Series Exam Plan.md, MAT700

now lets do the exact same thing for this second task. give me subtasks and everything.

Thought for 30s ›

Add these as **subtasks under the existing parent task**:

Parent task: `MM L1–L2 finish + Tutorial 1 P4–P5 TF-IDF`
Time: 14:45–16:30
Priority: P2
Labels: `core`, `lecture`, `exercise`
Reminder: none
Location: same as parent

Tiny adjustment: because this block is only **1h45**, don't try to "properly finish" all of Lecture 2's 67 slides. Treat L2 as a **targeted skim** for standard formulation + vector representation, then spend most of the block on P4–P5.

Subtasks to add in Todoist

1. Set up L1–L2 + Tutorial 1 P4–P5 workspace

Title:
`Set up L1-L2 and Tutorial 1 workspace`

Time: 14:45
Duration: 5 min
Priority: P4
Labels: `core`
Reminder: none
Location: leave blank / same as parent

Description:

```
Open:
- Lecture1.pdf
```

```
- Lecture2.pdf
- Tutorial1-beamer-version.pdf
- MA3700SolutionsTutorial1.pdf

Create one notebook page:
MM L1-L2 + Tutorial 1 P4-P5

Use sections:
1. L1-L2 quick notes
2. TF-IDF formula sheet
3. P4 full TF-IDF matrix
4. P5 vector space model
5. Mistakes / things to redo
```

---

2. Finish key L1–L2 concepts only

Title:
`Finish key L1-L2 concepts only`

Time: 14:50
Duration: 15 min
Priority: P2
Labels: `core`, `lecture`
Reminder: none
Location: leave blank / same as parent

Description:

```
Do a targeted skim, not a full deep read.

L1 quick check:
- data mining definition
- labelled vs unlabelled data
- classification vs regression
- why TF-IDF is useful for text/document data

L2 quick check:
- instance / record / attribute
- supervised vs unsupervised
- data as vectors
- document-term matrix as a way to represent text data numerically

Write only:
- definition
- why it matters
- possible exam trigger
```

```
Stop after 15 min. The main work today is P4-P5.
```

---

3. Make TF-IDF mini formula sheet

Title:

`Make TF-IDF mini formula sheet`

Time: 15:05

Duration: 10 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Write these formulas before starting P4:

IDF_i = log2(N / n_i)

TF_ij = f_ij / max_k(f_kj)

TF-IDF_ij = TF_ij × IDF_i

Where:
N = total number of documents
n_i = number of documents containing word i
f_ij = frequency of word i in document j
max_k(f_kj) = highest word frequency in document j

Important:
- IDF uses log base 2.
- Rare word → higher IDF.
- Common word → lower IDF.
- TF normalises by the most frequent word in the same document.
```

---

4. Do Tutorial 1 P4 full TF-IDF matrix

Title:

`Do Tutorial 1 P4 full TF-IDF matrix`

Time: 15:15

Duration: 30 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
Compute the full TF-IDF matrix from the given document corpus.

Steps:
1. Identify all terms/words in the corpus.
2. For each word, compute n_i:
   number of documents containing that word.
3. Compute IDF_i = log2(N / n_i) for each word.
4. For each document, find max_k(f_kj):
   the maximum frequency of any word in that document.
5. Compute TF_ij = f_ij / max_k(f_kj).
6. Compute TF-IDF_ij = TF_ij × IDF_i.
7. Put the results into a clear matrix:
   rows = words, columns = documents.

Checks:
- If a word does not appear in a document, TF-IDF = 0.
- If a word appears in every document, IDF should be low.
- Do not use ln or log10.
```

---

5. Do Tutorial 1 P5 vector space model

Title:

`Do Tutorial 1 P5 vector space model`

Time: 15:45

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
Convert the document-term / TF-IDF matrix into vector representation.

Steps:
1. Treat each document as a vector.
2. Each coordinate of the vector corresponds to one word/term.
3. The coordinate value is usually the TF-IDF score for that word in that document.
4. Write each document as:
   D1 = (score_word1, score_word2, score_word3, ...)
   D2 = (score_word1, score_word2, score_word3, ...)
   etc.

Interpretation:
Documents are no longer just text.
They become points/vectors in a numerical space.

Exam trigger:
Given a document-term matrix, write the vector representation of each document.
```

---

6. Check solutions and redo corrections

Title:

`Check P4-P5 solutions and redo corrections`

Time: 16:00

Duration: 18 min

Priority: P2

Labels: `core`, `exercise`, `review`

Reminder: none

Location: leave blank / same as parent

Description:

```
Open MA3700SolutionsTutorial1.pdf.

Check:
- P4 full TF-IDF matrix
- P5 vector space model

For every mistake:
1. Mark the exact wrong step.
2. Redo that calculation immediately.
3. Write why it was wrong.

Common mistakes to check:
- using wrong log base
```

```
- using N instead of n_i
- forgetting that TF is divided by max word frequency in that document
- treating missing words as blank instead of 0
- mixing rows and columns in the matrix
- writing vectors in inconsistent word order
```

---

7. Add error log + 3 exam flashcards

Title:

`Add error log and 3 exam flashcards`

Time: 16:18

Duration: 12 min

Priority: P4

Labels: `core`, `review`

Reminder: none

Location: leave blank / same as parent

Description:

```
Add mistakes to:
Mistakes / things to redo

Then create 3 flashcards:

1. What is IDF?
Answer:
IDF_i = log2(N / n_i), where N is total documents and n_i is documents containing

2. What is TF?
Answer:
TF_ij = f_ij / max_k(f_kj), frequency of word i in document j divided by the maxim

3. What is the vector space model?
Answer:
Represent each document as a numerical vector, usually using term weights such as T

Final check:
Can I explain P4 and P5 without looking at the solution?
```

Clean Todoist version

```
Parent task:
MM L1-L2 finish + Tutorial 1 P4-P5 TF-IDF
14:45-16:30 | P2 | @core @lecture @exercise

Subtasks:
1. Set up L1-L2 and Tutorial 1 workspace
   14:45 | 5 min | P4 | @core

2. Finish key L1-L2 concepts only
   14:50 | 15 min | P2 | @core @lecture

3. Make TF-IDF mini formula sheet
   15:05 | 10 min | P2 | @core @exercise

4. Do Tutorial 1 P4 full TF-IDF matrix
   15:15 | 30 min | P2 | @core @exercise

5. Do Tutorial 1 P5 vector space model
   15:45 | 15 min | P2 | @core @exercise

6. Check P4-P5 solutions and redo corrections
   16:00 | 18 min | P2 | @core @exercise @review

7. Add error log and 3 exam flashcards
   16:18 | 12 min | P4 | @core @review
```

Sources:

- Todoist Study Plan - Strict Structure.md, MAT700 Phase 1 task list: Fri 24 Apr task

perfect! now for this one

Thought for 18s ›

Absolutely — this one is **more proof/logic-heavy** than the TF-IDF block, so the structure should protect you from getting stuck too long on graph theory.

Add these as **subtasks under the existing parent task**:

Parent task: `MM L2-L3 + Tutorial 1 P6-P11`
Time: 14:45–16:30
Priority: P2
Labels: `core`, `lecture`, `exercise`
Reminder: none
Location: same as parent, Trevithick

Subtasks to add in Todoist

1. Set up L2-L3 and Tutorial 1 P6-P11 workspace

Title:
`Set up L2-L3 and Tutorial 1 workspace`

Time: 14:45
Duration: 5 min
Priority: P4
Labels: `core`
Reminder: none
Location: leave blank / same as parent

Description:

```
Open:
- Lecture2.pdf
- Lecture3.pdf
- Tutorial1-beamer-version.pdf
- MA3700SolutionsTutorial1.pdf

Create one notebook page:
MM L2-L3 + Tutorial 1 P6-P11

Use sections:
1. L2-L3 quick notes
2. Graph theory formulas
3. P6-P9 graph theory
4. P10-P11 Jaccard for sets
5. Mistakes / things to redo
```

---

2. Skim L2-L3 only for the tutorial topics

Title:

`Skim L2-L3 for graph and similarity topics`

Time: 14:50

Duration: 15 min

Priority: P2

Labels: `core`, `lecture`

Reminder: none

Location: leave blank / same as parent

Description:

```
Do a targeted skim, not a full detailed lecture read.

L2 focus:
- instance / record / attribute
- data as mathematical objects
- why data can be represented using tables, vectors, graphs, or sets

L3 focus:
- similar items problem
- replacing documents/items by sets
- Jaccard similarity for sets
- why intersection and union measure overlap

Write only:
- definition
- why it matters
- possible exam trigger

Stop after 15 min. Tutorial problems are the main work.
```

---

3. Make graph theory mini formula sheet

Title:

`Make graph theory mini formula sheet`

Time: 15:05

Duration: 10 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Write these before starting P6-P9:

For an undirected graph:
Sum of degrees of all vertices = 2|E|

Meaning:
Each edge contributes 2 to the total degree count,
because each edge touches two endpoints.

Adjacency matrix:
A_ij = 1 if there is an edge from i to j
A_ij = 0 otherwise

Important theorem:
The (i,j) entry of A^k counts the number of walks of length k
from vertex i to vertex j.

Exam trigger:
Given a graph, build A.
Given A^2 or A^k, interpret entries as numbers of walks.
```

---

4. Do Tutorial 1 P6: sum of degrees proof

Title:

`Do Tutorial 1 P6 sum of degrees proof`

Time: 15:15

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
Prove that the sum of all vertex degrees equals 2|E|.

Proof structure:
1. Let G = (V,E) be an undirected graph.
2. Degree of a vertex = number of edges incident to it.
3. If we add degrees over all vertices, each edge is counted twice:
   once at each endpoint.
4. Therefore:
```

```
   sum over vertices deg(v) = 2|E|.

Write it as a clean exam-style proof, not just one sentence.

Check:
- Did I explain why each edge is counted exactly twice?
- Did I use |E| correctly as the number of edges?
```

---

5. Do Tutorial 1 P7-P9 adjacency matrix powers

Title:

`Do Tutorial 1 P7-P9 adjacency matrix powers`

Time: 15:30

Duration: 25 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
Practise adjacency matrices and powers of adjacency matrices.

Steps:
1. Draw or inspect the given graph carefully.
2. Label vertices consistently.
3. Build the adjacency matrix A:
   row = starting vertex
   column = ending vertex
4. Compute the required matrix power:
   A^2, A^3, or whatever the problem asks.
5. Interpret entries:
   (A^k)_ij = number of walks of length k from i to j.

For proof-style question:
Explain why multiplying matrices counts possible intermediate vertices.

Example explanation:
A^2_ij = sum over r of A_ir A_rj.
Each nonzero product A_ir A_rj represents a walk i -> r -> j.

Checks:
- Is the vertex order consistent?
- For undirected graphs, is A symmetric?
```

```
- Are diagonal entries handled correctly?
- Did I say “walk”, not necessarily “path”?
```

---

6. Do Tutorial 1 P10-P11 Jaccard for sets

Title:

`Do Tutorial 1 P10-P11 Jaccard for sets`

Time: 15:55

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
Compute Jaccard similarity for sets.

Formula:
J(S,T) = |S ∩ T| / |S ∪ T|

Steps:
1. Write set S clearly.
2. Write set T clearly.
3. Find S ∩ T:
   elements appearing in both sets.
4. Find S ∪ T:
   elements appearing in either set.
5. Compute:
   |intersection| / |union|

Interpretation:
J = 0 means no overlap.
J = 1 means identical sets.
Higher J means more similarity.

Important:
This is Jaccard for SETS, not bags.
Do not count repeated elements here unless the question explicitly says bags.
```

---

7. Check solutions and redo corrections

Title:

`Check P6-P11 solutions and redo corrections`

Time: 16:10

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`, `review`

Reminder: none

Location: leave blank / same as parent

Description:

```
Open MA3700SolutionsTutorial1.pdf.

Check:
- P6 sum of degrees proof
- P7-P9 adjacency matrix / matrix powers
- P10-P11 Jaccard sets

For every mistake:
1. Mark the exact wrong step.
2. Redo that calculation or proof immediately.
3. Write why it was wrong.

Common mistakes:
- saying each edge is counted once instead of twice
- inconsistent vertex order in adjacency matrix
- mixing up rows and columns
- forgetting that A^k counts walks of length k
- confusing walk with path
- using Jaccard bag formula instead of set formula
- counting duplicate elements in sets
```

---

8. Add error log and 3 exam flashcards

Title:

`Add error log and 3 exam flashcards`

Time: 16:25

Duration: 5 min

Priority: P4

Labels: `core`, `review`

Reminder: none

Location: leave blank / same as parent

Description:

```
Add mistakes to:
Mistakes / things to redo

Create 3 flashcards:

1. What is the sum of degrees theorem?
Answer:
For an undirected graph, sum of all vertex degrees = 2|E|.

2. What does (A^k)_ij mean?
Answer:
The number of walks of length k from vertex i to vertex j.

3. What is Jaccard similarity for sets?
Answer:
J(S,T) = |S ∩ T| / |S ∪ T|.

Final check:
Can I explain P6, one adjacency matrix question, and one Jaccard question without l
```

Clean Todoist version

```
Parent task:
MM L2-L3 + Tutorial 1 P6-P11
14:45-16:30 | P2 | @core @lecture @exercise

Subtasks:
1. Set up L2-L3 and Tutorial 1 workspace
   14:45 | 5 min | P4 | @core

2. Skim L2-L3 for graph and similarity topics
   14:50 | 15 min | P2 | @core @lecture

3. Make graph theory mini formula sheet
   15:05 | 10 min | P2 | @core @exercise

4. Do Tutorial 1 P6 sum of degrees proof
   15:15 | 15 min | P2 | @core @exercise

5. Do Tutorial 1 P7-P9 adjacency matrix powers
   15:30 | 25 min | P2 | @core @exercise

6. Do Tutorial 1 P10-P11 Jaccard for sets
   15:55 | 15 min | P2 | @core @exercise

7. Check P6-P11 solutions and redo corrections
```

```
   16:10 | 15 min | P2 | @core @exercise @review

8. Add error log and 3 exam flashcards
   16:25 | 5 min | P4 | @core @review
```

Sources:

- Todoist Study Plan - Strict Structure.md, MAT700 Phase 1 task list: Sat 25 Apr task.
  Todoist Study Plan - Strict Str…
- Mathematical Methods for Data Mining and Time Series Exam Plan.md, Tutorial 1 overview: P6–P9 graph theory and P10–P11 Jaccard similarity.
  Mathematical Methods for Data M…
- Mathematical Methods for Data Mining and Time Series Exam Plan.md, MAT700



now this one

Thought for 18s ›

Yes — this one should be treated as a **proof-drill block**, not a normal exercise block. The goal is not just “understand the proof”; it is to be able to **write both proofs under exam pressure**.

Add these as **subtasks under the existing parent task**:

Parent task: `MM L4 Tutorial 2: Distance Measures P4–P5 proofs`
Time: 14:45–16:30
Priority: P2
Labels: `core`, `exercise`
Reminder: none
Location: same as parent, ASSL / Colum Drive

Subtasks to add in Todoist

1. Set up distance proof workspace

Title:

`Set up distance proof workspace`

Time: 14:45
Duration: 5 min
Priority: P4
Labels: `core`
Reminder: none
Location: leave blank / same as parent

Description:

```
Open:
- Lecture4.pdf
- Tutorial2-beamer-version.pdf
- MA3700SolutionsTutorial2.pdf

Create one notebook page:
MM L4 Tutorial 2 P4-P5 Distance Proofs

Use sections:
1. Distance axioms
2. P4 L-infinity proof
3. P5 edit distance proof
4. Timed proof attempts
5. Mistakes / things to redo
```

---

2. Memorise the 4 distance axioms

Title:

`Memorise the 4 distance axioms`

Time: 14:50
Duration: 10 min
Priority: P2
Labels: `core`, `exercise`
Reminder: none
Location: leave blank / same as parent

Description:

```
Write the 4 axioms from memory.

For a function d(x,y) to be a distance measure:

1. Non-negativity:
   d(x,y) ≥ 0

2. Identity of indiscernibles:
   d(x,y) = 0 if and only if x = y

3. Symmetry:
   d(x,y) = d(y,x)

4. Triangle inequality:
   d(x,z) ≤ d(x,y) + d(y,z)

Important:
For P4 and P5, every proof must check ALL 4 axioms.
Triangle inequality is usually the hardest and most important one.
```

---

3. Build the L-infinity proof skeleton

Title:

`Build L-infinity proof skeleton`

Time: 15:00

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
Prepare the proof that L∞ is a distance measure.

Definition:
d∞(x,y) = max_i |x_i - y_i|

Proof skeleton:

1. Non-negativity:
Absolute values are non-negative, so the maximum is non-negative.

2. Identity:
```

```
d∞(x,y) = 0 means max_i |x_i-y_i| = 0.
So every |x_i-y_i| = 0.
Therefore x_i = y_i for all i, so x = y.
Also if x = y, then d∞(x,y) = 0.

3. Symmetry:
|x_i-y_i| = |y_i-x_i| for every coordinate.
So d∞(x,y) = d∞(y,x).

4. Triangle inequality:
For every i:
|x_i-z_i| ≤ |x_i-y_i| + |y_i-z_i|

So:
max_i |x_i-z_i|
≤ max_i (|x_i-y_i| + |y_i-z_i|)
≤ max_i |x_i-y_i| + max_i |y_i-z_i|

Therefore:
d∞(x,z) ≤ d∞(x,y) + d∞(y,z).

Final check:
Can I explain why max(a_i + b_i) ≤ max(a_i) + max(b_i)?
```

---

4. Write P4 L-infinity proof properly

Title:

`Write P4 L-infinity proof properly`

Time: 15:15

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Now write the full P4 proof in clean exam style.

Rules:
- Do not just write bullet points.
- Use mathematical sentences.
- Check all 4 axioms.
- Spend most care on triangle inequality.

After writing:
```

```
Mark the proof out of 4:
1 mark = non-negativity
1 mark = identity
1 mark = symmetry
1 mark = triangle inequality

If one axiom is weak, rewrite that part immediately.
```

---

5. Build the edit distance proof skeleton

Title:

`Build edit distance proof skeleton`

Time: 15:30

Duration: 20 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
Prepare the proof that edit distance is a distance measure.

Definition:
Edit distance d(x,y) = the smallest number of insertions and deletions needed to tr

Proof skeleton:

1. Non-negativity:
d(x,y) counts a number of operations.
A number of operations cannot be negative.

2. Identity:
If x = y, no operation is needed, so d(x,y) = 0.
If d(x,y) = 0, then x can be changed into y with no operations.
Therefore x = y.

3. Symmetry:
If x can be changed into y using k insertions/deletions,
then y can be changed back into x using the reverse operations.
Insertions become deletions and deletions become insertions.
So d(x,y) = d(y,x).

4. Triangle inequality:
Suppose d(x,y) = a and d(y,z) = b.
```

```
There is a sequence of a operations from x to y.
There is a sequence of b operations from y to z.
Joining them gives a sequence of a+b operations from x to z.
But d(x,z) is the smallest possible number of operations from x to z.
So:
d(x,z) ≤ a+b = d(x,y)+d(y,z).

Key phrase:
“Smallest possible” is the reason triangle inequality works.
```

---

6. Write P5 edit distance proof properly

Title:

`Write P5 edit distance proof properly`

Time: 15:50
Duration: 15 min
Priority: P2
Labels: `core`, `exercise`
Reminder: none
Location: leave blank / same as parent

Description:

```
Now write the full P5 proof in clean exam style.

Rules:
- State the definition first.
- Check all 4 axioms.
- For symmetry, explicitly mention reversing operations.
- For triangle inequality, explicitly mention concatenating two edit sequences.
- Use the phrase “smallest number of operations”.

After writing:
Mark the proof out of 4:
1 mark = non-negativity
1 mark = identity
1 mark = symmetry
1 mark = triangle inequality

If the triangle inequality is unclear, rewrite only that part.
```

---

7. Check solutions and compare proof style

Title:

`Check P4-P5 solutions and compare proof style`

Time: 16:05

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`, `review`

Reminder: none

Location: leave blank / same as parent

Description:

```
Open MA3700SolutionsTutorial2.pdf.

Check:
- P4 L∞ proof
- P5 edit distance proof

For each proof, compare:
1. Did I state the definition?
2. Did I check all 4 axioms?
3. Did I prove both directions of d(x,y)=0 iff x=y?
4. Did I make triangle inequality clear?
5. Did I use correct notation?

Common mistakes:
- forgetting the “only if” direction in axiom 2
- saying triangle inequality is obvious without proving it
- not explaining max_i properly in L∞
- forgetting that edit distance is the smallest number of operations
- confusing edit distance operations with substitution if the tutorial only allows
```

---

8. Timed proof drill: 3 minutes each

Title:

`Timed proof drill 3 minutes each`

Time: 16:20

Duration: 7 min

Priority: P3

Labels: `core`, `exercise`, `review`

Reminder: none

Location: leave blank / same as parent

Description:

```
Close notes.

Timed attempt:
- 3 min: write the L∞ proof skeleton from memory.
- 3 min: write the edit distance proof skeleton from memory.
- 1 min: circle missing axioms or weak parts.

Goal:
Not perfect writing.
Just prove that I can recall the structure under time pressure.
```

---

9. Add error log and proof flashcards

Title:

`Add error log and proof flashcards`

Time: 16:27

Duration: 3 min

Priority: P4

Labels: `core`, `review`

Reminder: none

Location: leave blank / same as parent

Description:

```
Add mistakes to:
Mistakes / things to redo

Create 3 flashcards:

1. What are the 4 distance axioms?
Answer:
Non-negativity, identity, symmetry, triangle inequality.

2. What is the key step in the L∞ triangle inequality?
Answer:
Use |x_i-z_i| ≤ |x_i-y_i| + |y_i-z_i| for each coordinate, then take max over i.

3. What is the key step in the edit distance triangle inequality?
Answer:
Join an optimal path from x to y with an optimal path from y to z. This gives a val
```

Clean Todoist version

```
Parent task:
MM L4 Tutorial 2: Distance Measures P4-P5 proofs
14:45-16:30 | P2 | @core @exercise

Subtasks:
1. Set up distance proof workspace
   14:45 | 5 min | P4 | @core

2. Memorise the 4 distance axioms
   14:50 | 10 min | P2 | @core @exercise

3. Build the L-infinity proof skeleton
   15:00 | 15 min | P2 | @core @exercise

4. Write P4 L-infinity proof properly
   15:15 | 15 min | P2 | @core @exercise

5. Build the edit distance proof skeleton
   15:30 | 20 min | P2 | @core @exercise

6. Write P5 edit distance proof properly
   15:50 | 15 min | P2 | @core @exercise

7. Check P4-P5 solutions and compare proof style
   16:05 | 15 min | P2 | @core @exercise @review

8. Timed proof drill 3 minutes each
   16:20 | 7 min | P3 | @core @exercise @review

9. Add error log and proof flashcards
   16:27 | 3 min | P4 | @core @review
```

Sources:

- `Todoist Study Plan - Strict Structure.md`, MAT700 Phase 1 task list: Mon 27 Apr distance proofs task. Todoist Study Plan - Strict Str...
- `Mathematical Methods for Data Mining and Time Series Exam Plan.md`, MAT700 Tutorial 2 overview and distance-proof revision priorities. Mathematical Methods for Data M...
- `5 - MAT700MA4700 Mathematical Methods for Data Mining online class-20260216_151041-Meeting Recording.txt`, distance axioms and edit-distance proof explanation. 5 - MAT700MA4700 Mathematical M...

now this one?

Thought for 22s ›

Yes — this one is a **calculation + matrix setup block**. The danger is spending too long reading the PageRank lecture and not actually doing the matrix iterations. So keep the lecture reading targeted.

Add these as **subtasks under the existing parent task**:

Parent task: `MM L5 PageRank + Tutorial 3 Link Analysis P1-P2`
Time: 15:00–16:30
Priority: P2
Labels: `core`, `lecture`, `exercise`
Reminder: none
Location: same as parent, Abacws

Subtasks to add in Todoist

1. Set up PageRank workspace

Title:
`Set up PageRank workspace`

Time: 15:00
Duration: 5 min
Priority: P4
Labels: `core`
Reminder: none
Location: leave blank / same as parent

Description:

```
Open:
- Lecture5.pdf
- Tutorial3.pdf
- Tutorial3_Solutions.pdf

Create one notebook page:
MM L5 PageRank + Tutorial 3 Link Analysis P1-P2

Use sections:
1. PageRank formula sheet
2. Transition matrix construction
3. Tutorial 3 Link Analysis P1
4. Tutorial 3 Link Analysis P2
5. Mistakes / things to redo
```

---

2. Read L5 only for PageRank essentials

Title:

`Read L5 PageRank essentials only`

Time: 15:05

Duration: 20 min

Priority: P2

Labels: `core`, `lecture`

Reminder: none

Location: leave blank / same as parent

Description:

```
Do a targeted read, not a full slow read of all 54 slides.

Focus only on:
- web graph idea
- pages as nodes
- links as directed edges
- transition matrix M
- column-stochastic matrix
- random surfer idea
- damping factor beta
- PageRank update formula

Write only:
- definition
- why it matters
- possible exam trigger
```

```
Stop after 20 min.
The main work is doing Tutorial 3 Link Analysis P1-P2.
```

---

3. Make PageRank mini formula sheet

Title:

`Make PageRank mini formula sheet`

Time: 15:25

Duration: 10 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Write these formulas before starting the tutorial:

PageRank update:
v_new = (1 - beta)e/n + beta Mv

Where:
v = PageRank vector
beta = damping factor
e = vector of all 1s
n = number of pages
M = transition matrix

Column-stochastic rule:
Each column of M must sum to 1.

How to build M:
Column j represents links going OUT from page j.
If page j links to k pages, each linked page gets 1/k in column j.

Checks:
- Are all entries non-negative?
- Does every column sum to 1?
- Does the PageRank vector sum to 1 after each iteration?
```

---

4. Build transition matrix carefully

Title:

`Build transition matrix carefully`

Time: 15:35

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
For Tutorial 3 Link Analysis P1-P2, build the transition matrix M correctly.

Steps:
1. Label pages/nodes consistently.
2. For each page, list its outgoing links.
3. For each page j, fill column j of M.
4. If page j has k outgoing links, put 1/k in the rows it links to.
5. Put 0 for pages it does not link to.
6. Check every column sums to 1.

Important:
Do NOT make M row-stochastic unless the notes/problem explicitly use row convention
The task says column-stochastic transition matrix.

Exam trigger:
Most PageRank mistakes happen before the iteration starts, because M is built the w
```

---

5. Do Tutorial 3 Link Analysis P1

Title:

`Do Tutorial 3 Link Analysis P1`

Time: 15:50

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
Compute PageRank for the first small web graph.

Steps:
1. Build M.
2. Check columns sum to 1.
3. Set starting vector v0.
   Usually use equal probabilities unless the question gives another vector.
4. Apply:
   v_new = (1 - beta)e/n + beta Mv
5. Do the number of iterations requested.
6. After each iteration, check:
   entries are non-negative
   entries sum to 1
7. Rank pages from highest PageRank to lowest.

If the tutorial asks for exact PageRank instead of iterations:
Set v = (1 - beta)e/n + beta Mv
and solve the system together with sum(v_i)=1.
```

---

6. Do Tutorial 3 Link Analysis P2

Title:

`Do Tutorial 3 Link Analysis P2`

Time: 16:05

Duration: 15 min

Priority: P2

Labels: `core`, `exercise`

Reminder: none

Location: leave blank / same as parent

Description:

```
Goal:
Repeat the PageRank process for P2, but faster and cleaner.

Steps:
1. Draw/list the graph.
2. Build M column by column.
3. Check every column sum.
4. Apply PageRank update:
   v_new = (1 - beta)e/n + beta Mv
5. Do the required iterations or solve exact equation if asked.
6. Write final ranking clearly.
```

```
Common checks:
- Did I divide by the number of outgoing links?
- Did I put probabilities in the correct column?
- Did I include the teleportation term (1-beta)e/n?
- Does the final vector sum to 1?
```

---

7. Check solutions and fix matrix errors

Title:

`Check PageRank solutions and fix matrix errors`

Time: 16:20

Duration: 7 min

Priority: P2

Labels: `core`, `exercise`, `review`

Reminder: none

Location: leave blank / same as parent

Description:

```
Open Tutorial3_Solutions.pdf.

Check:
- P1 transition matrix
- P1 PageRank calculation
- P2 transition matrix
- P2 PageRank calculation

For every mistake:
1. Mark whether it was:
   - graph reading error
   - wrong row/column convention
   - wrong outgoing-link division
   - arithmetic error
   - forgot teleportation term
2. Redo only the wrong step.
3. Add the mistake to the error log.
```

---

8. Add error log and PageRank flashcards

Title:

`Add error log and PageRank flashcards`

Time: 16:27

Duration: 3 min

Priority: P4

Labels: `core`, `review`

Reminder: none

Location: leave blank / same as parent

Description:

```
Add mistakes to:
Mistakes / things to redo

Create 3 flashcards:

1. What is the PageRank update formula?
Answer:
v_new = (1 - beta)e/n + beta Mv.

2. What does column-stochastic mean?
Answer:
Each column of the transition matrix sums to 1.

3. How do I fill column j of M?
Answer:
Look at outgoing links from page j. If page j links to k pages, each linked page ge

Final check:
Can I build M without looking at the solution?
Can I explain why the PageRank vector should sum to 1?
```

Clean Todoist version

```
Parent task:
MM L5 PageRank + Tutorial 3 Link Analysis P1-P2
15:00-16:30 | P2 | @core @lecture @exercise

Subtasks:
1. Set up PageRank workspace
   15:00 | 5 min | P4 | @core

2. Read L5 PageRank essentials only
   15:05 | 20 min | P2 | @core @lecture

3. Make PageRank mini formula sheet
   15:25 | 10 min | P2 | @core @exercise
```

```
4. Build transition matrix carefully
   15:35 | 15 min | P2 | @core @exercise

5. Do Tutorial 3 Link Analysis P1
   15:50 | 15 min | P2 | @core @exercise

6. Do Tutorial 3 Link Analysis P2
   16:05 | 15 min | P2 | @core @exercise

7. Check PageRank solutions and fix matrix errors
   16:20 | 7 min | P2 | @core @exercise @review

8. Add error log and PageRank flashcards
   16:27 | 3 min | P4 | @core @review
```

Sources:

- `Todoist Study Plan - Strict Structure.md`, MAT700 Phase 1 task: “MM L5 PageRank + Tutorial 3 Link Analysis P1–P2”. Todoist Study Plan - Strict Str…
- `Mathematical Methods for Data Mining and Time Series Exam Plan.md`, Tutorial 3 Link Analysis and PageRank formula notes. Mathematical Methods for Data M…

I only have one day left and I'm just starting lecture 2. I have lecture 3, 4, 5, 6, 7 left. Can you give me an hour? So I'm gonna wake up tomorrow at 8 and be ready at 8:30 to start studying and I will be studying till midnight till 2 a.m. And then the day after that, I'm gonna wake up at 8:30 again or even 7:30 and study till 11 and 11 will go to take my exam. I want to finish reading all of these slides and lectures so soon to be able to do reviewing and, you know, formula memorizing, exam questions, reading from the question banks, etc. Help me with that.

Thought for 33s ›

Answer

Zahra, with **one day left**, the plan cannot be “read all slides properly, then review.” That will trap you.

The survival strategy is:

```
Slides = fast topic exposure
Past papers = what the exam actually asks
Question banks = drilling
```

```
Formula sheet = memory anchor
Tutorials = worked examples
```

Your exam pattern is very consistent: **4 questions, answer 3, 2 hours, 75 marks**, and the recent papers allow **one A4 sheet, 2 sides, of your own notes**. 2024 and 2025 also say calculators are permitted, but 2023 says calculators were not permitted, so take your calculator but do not rely on it completely. MAT700 Mathematical Methods for… MAT700 Mathematical Methods for… MAT700 Mathematical Methods for…

Your highest-yield topics are absolutely clear from 2023–2025:

```
TF-IDF
Bonferroni / birthday expectation
graph adjacency + sum of degrees
distance proofs: L∞, edit, Jaccard distance
Jaccard sets + bags
kNN weighted/unweighted
Naive Bayes
precision/recall/F-measure
PageRank
k-means
halving / weighted majority
softmax + cross entropy
mathematical framework for learning
```

2025 alone asks TF-IDF, document-term matrix, adjacency matrix, L∞ proof, edit proof, Jaccard distance proof, Naive Bayes, learning framework, kNN, softmax/cross entropy, k-means, PageRank/spider traps, F-measure, halving and weighted majority. MAT700 Mathematical Methods for…

Tomorrow plan: 08:30 to midnight

08:30–08:45 — Set up command centre

Goal: no hunting for files later.

Open:

```
Lecture2.pdf
Lecture3.pdf
Lecture4.pdf
Lecture5.pdf
Lecture6.pdf
Lecture7.pdf
```

```
MAT700 2023 paper
MAT700 2024 paper
MAT700 2025 paper

Formula cheat sheet / blank A4 notes sheet
Question banks
Study notes
Tutorial solutions
```

Create one page called:

```
MAT700 Final 24h Error Log
```

Sections:

```
1. Formulas I forgot
2. Proofs I cannot write
3. Calculation mistakes
4. Topics to revisit on exam morning
```

08:45–10:15 — Lecture 2: TF-IDF, learning framework, graphs

Do not read all 67 slides slowly.

Study only these exam-trigger topics:

```
- supervised vs unsupervised learning
- instance-based vs model-based learning
- mathematical framework for learning
- hypothesis space
- loss function
- expected loss
- TF-IDF
- document-term matrix
- graph basics / adjacency matrix
- softmax and cross entropy
```

Then immediately do these from past papers:

```
2024 Q1(b): mathematical framework for learning
2024 Q1(c): softmax + cross entropy
2024 Q2(a-c): TF-IDF
2025 Q1(b): TF-IDF + document-term matrix
2025 Q1(c): adjacency matrix + sum of degrees
2025 Q2(e): mathematical framework
2025 Q3(b): softmax + cross entropy
```

Output by 10:15:

```
- one clean definition of TF-IDF
- one clean definition of document-term matrix
- one clean paragraph for mathematical framework
- one formula for softmax
- one formula for cross entropy
- proof skeleton for sum of degrees = 2|E|
```

Lecture 2 maps strongly to standard formulation, TF-IDF, document-term matrix and graph basics in your mapping file. Tutorials and Transcriptions Ma…

10:15–10:30 — Break

No phone scrolling. Tea, water, stretch.

10:30–11:45 — Lecture 3: Jaccard, shingling, minhashing

Focus only on:

```
- similar items problem
- Jaccard similarity for sets
- Jaccard similarity for bags
- Jaccard distance
- shingling
- minhash idea
```

Do these:

```
2024 Q2(e): Jaccard sets
2024 Q2(f): Jaccard bags
2024 Q2(g): prove Jaccard similarity of bags ≤ 1/2
2024 Q2(h): Jaccard distance proof
2025 Q2(c): Jaccard distance proof
```

Output by 11:45:

```
Jaccard set formula:
J(S,T) = |S ∩ T| / |S ∪ T|

Jaccard bag formula:
J(B,C) = sum min(counts) / sum max(counts)
```

```
Jaccard distance:
d(S,T) = 1 - J(S,T)
```

Lecture 3 is the similar-items/Jaccard/shingling/minhashing lecture, and Tutorial 1 P10–P11 plus Tutorial 2 Jaccard bags map to it. Tutorials and Transcriptions Ma…

11:45–12:30 — Lunch + passive review

While eating, only listen/read lightly:

```
- Jaccard formulas
- TF-IDF formula
- distance axioms
```

No new topic.

12:30–14:30 — Lecture 4: the most important block

This is the biggest marks block. Focus on:

```
- 4 distance axioms
- L∞ proof
- edit distance proof
- Jaccard distance proof
- kNN
- weighted kNN
- precision, recall, F-measure
- F ≤ (P+R)/2 proof
- Naive Bayes
```

Do these:

```
2023 Q1(d): edit distance proof
2023 Q2(a): precision/recall/F-measure
2023 Q3(a–c): kNN + Naive Bayes
2024 Q2(d): edit distance proof
2024 Q3(e–f): F-measure
2024 Q4(b–c): kNN + Naive Bayes
2025 Q2(a–d): L∞, edit, Jaccard distance, Naive Bayes
2025 Q3(a): kNN
2025 Q4(a): F-measure
```

Output by 14:30:

You must be able to write these **without notes**:

```
1. 4 distance axioms
2. L∞ distance proof
3. edit distance proof
4. Jaccard distance proof outline
5. precision / recall / F formula
6. Naive Bayes formula
7. kNN majority + weighted voting explanation
```

Lecture 4 is mapped to distances, kNN, evaluation, precision/recall/F-measure and Naive Bayes. Tutorials and Transcriptions Ma…

14:30–14:45 — Break

Walk. Water. Do not “just check something” online.

14:45–16:00 — Lecture 5: PageRank

Focus only on:

```
- web graph
- transition matrix M
- column-stochastic matrix
- random surfer
- damping factor β
- PageRank update formula
- spider traps
```

Do:

```
2023 Q2(b): PageRank description/proof-style question
2024 Q3(d): compute PageRank with β = 0.8
2025 Q3(d): spider traps
2025 Q4(d): compute PageRank with β = 0.8
```

Output by 16:00:

```
PageRank formula:
v = ((1 - β)e/n) + βMv

Rules:
- M is column-stochastic.
- Column j distributes page j's rank to outgoing links.
- Spider trap problem is reduced by teleportation term.
```

Lecture 5 maps directly to PageRank, transition matrices, random surfer interpretation, spider traps and PageRank calculation. Tutorials and Transcriptions Ma…

16:00–16:15 — Break

Eat something small. Your brain will start resisting here.

16:15–17:30 — Lecture 6: clustering and k-means

Focus only on:

```
- clustering definition
- hierarchical clustering
- agglomerative/bottom-up idea
- distance between clusters
- k-means algorithm
- k-means objective function
- convergence idea
- initial centroid selection
```

Do:

```
2023 Q3(d): k-means algorithm + objective
2024 Q4(d): k-means algorithm + objective + convergence
2025 Q3(c): k-means + objective + initial centroids
```

Output by 17:30:

```
K-means algorithm:
1. Choose k initial centroids.
2. Assign each point to nearest centroid.
3. Recompute centroids as means of assigned points.
4. Repeat until assignments/centroids stop changing.

Objective:
minimise sum of squared distances from points to their assigned centroids.
```

Lecture 6 maps to clustering and k-means, including hierarchical clustering, centroids, objective function, convergence and initial centroids. Tutorials and Transcriptions Ma…

17:30–18:30 — Lecture 7: halving, weighted majority, AdaBoost

Focus only on:

```
- meta-algorithm
- online learning
- experts
- halving algorithm
- proof of at most log2(n) mistakes
- weighted majority algorithm
- AdaBoost idea
```

Do:

```
2023 Q4(c): meta-algorithm + halving proof
2024 Q3(c): halving / weighted majority
2024 Q4(a): halving proof
2025 Q4(b–c): meta-algorithm + halving proof
```

Output by 18:30:

```
Halving proof skeleton:
- Start with n experts.
- At each mistake, at least half of remaining experts are removed.
- After m mistakes, at most n / 2^m experts remain.
- Since at least one consistent expert remains, n / 2^m ≥ 1.
- Therefore 2^m ≤ n, so m ≤ log2(n).
```

Lecture 7 maps to meta-algorithms, online learning, halving, weighted majority and AdaBoost idea. Tutorials and Transcriptions Ma…

18:30–19:15 — Dinner

No hard maths. Let memory settle.

Evening: stop “learning”, start exam training

19:15–20:45 — 2025 paper: planning drill

Do not fully solve everything. Do this:

```
Open MAT700 2025.
For every subquestion:
- identify topic
- write formula/proof skeleton
- mark: can answer / shaky / cannot answer
```

Then choose the **best 3 questions** you would answer in the real exam.

This matters because the exam asks you to answer **three out of four**, so selection strategy is part of the exam. MAT700 Mathematical Methods for...

20:45–21:45 — Build your A4 formula/notes sheet

Use this structure:

Front side

```
1. TF-IDF
2. Bonferroni / birthday expectation
3. Jaccard sets and bags
4. distance axioms
5. L∞ / edit / Jaccard proof keywords
6. precision / recall / F
```

Back side

```
7. kNN
8. Naive Bayes
9. PageRank
10. k-means
11. softmax / cross entropy
12. halving / weighted majority
```

Do not make it pretty. Make it dense and usable.

21:45–22:00 — Break

Short break. Then final hard drill.

22:00–23:15 — Timed micro-drills

Set a timer and do these closed-book:

```
8 min: TF-IDF calculation
6 min: birthday expectation
5 min: L∞ proof
5 min: edit distance proof
6 min: Jaccard set/bag calculation
8 min: kNN weighted vote
8 min: Naive Bayes posterior
```

```
10 min: PageRank setup
5 min: halving proof
```

The point is not perfection. The point is to stop freezing.

23:15–00:00 — Final memory lock

Read your A4 sheet three times.

Then from memory write:

```
- TF-IDF formula
- PageRank formula
- Naive Bayes formula
- precision/recall/F
- k-means objective
- 4 distance axioms
- halving proof skeleton
```

Stop at midnight if you can. If anxiety is too high, you may do **one extra 30-minute light review**, but do not start new topics at 1 a.m.

Exam morning plan

Assuming you need to leave/go to the exam around **11:00**:

07:30–08:00 — Wake, food, water

Protein + carbs. No heavy coffee panic.

08:00–08:45 — Proof recitation

Write from memory:

```
1. L∞ proof
2. edit distance proof
3. Jaccard distance proof
4. halving proof
5. F-measure inequality proof
```

08:45–09:30 — Formula pass

Read and rewrite:

```
TF-IDF
Jaccard bags
Naive Bayes
PageRank
k-means
softmax/cross entropy
precision/recall/F
```

09:30–10:15 — One mini calculation each

Do tiny versions only:

```
1 TF-IDF
1 kNN
1 Naive Bayes setup
1 PageRank matrix setup
```

10:15–10:45 — Exam strategy

Write this on scrap before leaving:

```
Read all 4 questions first.
Pick best 3.
Start with formula/definition marks.
For proofs: list all axioms, then prove.
For calculations: write setup even if arithmetic is unfinished.
Never spend more than 35–38 min on one question.
```

10:45–11:00 — Stop

Pack:

```
student ID
pens
calculator
water
A4 notes sheet if allowed
```

Then go.

What to skip

Skip these unless they appear directly in your question bank or past-paper answers:

```
- MNIST notebook details
- long transcript reading
- beautiful full summaries
- copying slides
- deep shingling/minhash proofs
- trying to master every single slide
```

The mapping says MNIST is mostly conceptual support, not primary written-exam practice.

Tutorials and Transcriptions Ma…

Your absolute minimum checklist

Before sleeping, you need to be able to answer these:

```
1. Explain supervised vs unsupervised learning.
2. Explain the mathematical framework for learning.
3. Define TF-IDF and compute it.
4. Define document-term matrix.
5. Compute birthday expectation.
6. Compute Jaccard for sets and bags.
7. Prove L∞ is a distance.
8. Prove edit distance is a distance.
9. Prove/outline Jaccard distance is a distance.
10. Do kNN majority and weighted voting.
11. Compute precision, recall, F-measure.
12. Do Naive Bayes classification.
13. Build PageRank matrix and use formula.
14. Explain spider traps.
15. State k-means algorithm and objective.
16. Prove halving mistakes ≤ log2(n).
17. Define softmax and cross entropy.
```

That is your exam survival set.

Sources

- `MAT700 Mathematical Methods for Data 2025.pdf`, p.1–4.
  MAT700 Mathematical Methods for…
- `MAT700 Mathematical Methods for Data 2024.pdf`, p.1–4.
  MAT700 Mathematical Methods for…

Okay, I fucked up again. Because I studied pretty late last night because I couldn't sleep, I woke up at 1, started lecture 2, and now I am on page 10 of 37. It's 3:25. I've been taking some time to make myself breakfast and lunch as well, based on my feed and how I should let this feed off. Can you give me a new plan? I'm on Formalization Model-Based Learning, Formalization of Model-Based Learning. That's where I am in lecture 2. Give me a new plan, please, based on the fuck-up. Thank you.
I'll probably won't be able to sleep till 4 a.m. like last night out of stress, so it's fine if you put like a plan for me to study

Show more

Thought for 14s

Answer

Zahra, this is still recoverable, but the plan must change completely:

No more “proper reading”.
From now on:

```
Study notes/question banks > full slides
Past-paper repeated topics > rare details
Proof skeletons + formulas > beautiful understanding
Closed-book recall > highlighting
```

The exam is **4 questions, answer 3**, so your goal is not to know everything. Your goal is to make **at least 3 questions answerable**. Recent papers repeatedly hit TF-IDF, graph basics, distance proofs, Jaccard, kNN, Naive Bayes, PageRank, k-means, halving/weighted majority, softmax/cross entropy, and precision/recall/F-measure.

MAT700 Mathematical Methods for… MAT700 Mathematical Methods for… MAT700 Mathematical Methods for…

New plan from 15:25 today

15:25–16:05 — Finish Lecture 2 only as exam skeleton

You are on **Formalisation of Model-Based Learning**. Do **not** read every sentence.

For Lecture 2, extract only:

```
instance-based vs model-based learning
mathematical framework for learning
hypothesis space
loss function
expected loss
empirical mean
softmax
cross entropy
TF-IDF
document-term matrix
adjacency matrix
sum of degrees = 2|E|
```

The lecture recording emphasises that model-based learning is about finding a function/model from data, choosing a hypothesis space, defining a loss function, and minimising expected loss; when the true distribution is unknown, expected value is approximated by empirical mean. 3 - MAT700MA4700 Mathematical M...

Hard stop at 16:05. Even if Lecture 2 feels unfinished.

16:05–16:45 — Lecture 2 question bank / past-paper hits

Do only these:

```
TF-IDF formula + 1 example
document-term matrix definition
mathematical framework paragraph
softmax + cross entropy definitions
sum of degrees = 2|E| proof skeleton
```

Write them into your formula/A4 sheet immediately.

16:45–17:30 — Lecture 3 crash: Jaccard only

Skip deep minhashing unless it is in your generated notes as a tiny definition.

Learn:

```
Jaccard sets
Jaccard bags
Jaccard distance
shingling definition
why similar documents become similar sets
proof that bag similarity ≤ 1/2
```

Lecture 3 is mapped to similar items, Jaccard similarity, Jaccard bags, shingling, minhashing, and near-duplicate documents; its tutorial mapping is Tutorial 1 P10–P11 and Tutorial 2 Jaccard bags P1.

MAT700 lecture mapping

17:30–18:00 — Eat / break

No studying while eating except maybe looking at the formula sheet.

18:00–19:45 — Lecture 4: highest-value block

This is your **most important block**.

Learn these in order:

```
4 distance axioms
L∞ proof
edit distance proof
Jaccard distance proof
kNN majority
weighted kNN
precision / recall / F-measure
Naive Bayes
```

The Lecture 4 transcript defines distance as needing non-negativity, identity, symmetry, and triangle inequality; it also covers Jaccard distance, edit distance, Hamming distance, and kNN. 5 - MAT700MA4700 Mathematical M…

Do not try to understand every proof perfectly. Memorise the structure.

19:45–20:30 — Lecture 4 drills

Closed-book, write:

```
1. Four distance axioms
2. L∞ proof skeleton
3. Edit distance proof skeleton
4. Jaccard distance proof skeleton
5. Precision / recall / F formula
6. Naive Bayes formula
7. Weighted kNN rule
```

If one is weak, mark it with a star. Do not spiral. Move on.

20:30–21:15 — Lecture 5: PageRank

Learn:

```
directed web graph
transition matrix M
column-stochastic matrix
random surfer
damping factor β
PageRank formula
spider traps
```

Use this exam answer:

```
PageRank assigns an importance score to each page.
A page is important if important pages link to it.
The transition matrix M describes movement along links.
The damping/teleportation term prevents rank getting trapped in spider traps.
```

Lecture 5 is mapped to PageRank, transition matrices, random surfer, damping factor, dead ends/spider traps, and Tutorial 3 Link Analysis P1–P2.

MAT700 lecture mapping

21:15–22:00 — PageRank drill

Do not spend forever calculating.

Practise:

```
How to build M:
Column j = where page j points.
If page j has r outgoing links, each linked page gets 1/r in column j.

Formula:
v = ((1 - β)e/n) + βMv
```

Then read one worked PageRank example from the question bank.

22:00–22:45 — Lecture 6: k-means only

Skip long clustering theory unless you have energy.

Learn:

```
clustering definition
hierarchical vs point-assignment idea
k-means algorithm
objective function
initial centroid selection
local minimum / convergence idea
```

The Lecture 6 recording frames clustering as grouping objects by similarity/distance, distinguishes hierarchical clustering from point-assignment approaches, and introduces k-means in Euclidean space with fixed k. 7 - MAT700MA4700 Mathematical M...

Exam paragraph:

```
K-means chooses k initial centroids, assigns each point to its nearest centroid, re
```

22:45–23:30 — Lecture 7: halving / weighted majority

Learn only:

```
meta-algorithm
experts
halving algorithm
weighted majority algorithm
halving proof: mistakes ≤ log2(n)
AdaBoost one-line idea
```

Memorise this proof:

```
Start with n experts.
Whenever the algorithm makes a mistake, at least half of the remaining experts are
Remove wrong experts.
After m mistakes, at most n / 2^m experts remain.
Since one expert is consistent, at least one expert remains.
So n / 2^m ≥ 1.
Therefore 2^m ≤ n.
So m ≤ log2(n).
```

Lecture 7 is mapped to meta-algorithms, AdaBoost, online learning, experts, halving, and weighted majority.

MAT700 lecture mapping

23:30–00:30 — Build the A4 sheet

Do this now, not at 3 a.m.

Front side

```
Bonferroni / birthday expectation
TF-IDF
document-term matrix
graph adjacency matrix
sum deg(v)=2|E|
Jaccard set + bag
distance axioms
L∞ / edit / Jaccard proof keywords
```

Back side

```
kNN + weighted kNN
precision / recall / F
Naive Bayes
softmax / cross entropy
PageRank
k-means
halving / weighted majority
```

Your past papers explicitly allow **one A4 sheet, two sides, of your own notes**.

MAT700 Mathematical Methods for… MAT700 Mathematical Methods for… MAT700 Mathematical Methods for…

00:30–01:45 — 2025 paper only

Open 2025. For every subquestion, write beside it:

```
know / shaky / skip
```

Then practise only the “shaky but possible” ones.

The 2025 paper is the best final guide because it contains: Bonferroni, TF-IDF, document-term matrix, adjacency matrix, L∞ proof, edit proof, Jaccard distance proof, Naive Bayes, mathematical framework, kNN, softmax/cross entropy, k-means, PageRank/spider traps, F-measure, halving, and weighted majority. MAT700 Mathematical Methods for…

01:45–02:30 — Closed-book memory dump

On blank paper, write from memory:

```
TF-IDF
Jaccard set/bag
4 distance axioms
edit distance proof
L∞ proof
Naive Bayes
precision/recall/F
PageRank
k-means algorithm
halving proof
```

Then compare with the A4 sheet.

02:30–03:15 — Error-log repair

Only fix the top 5 weaknesses.

Not 20. Not everything.

Choose from:

```
TF-IDF calculation
Jaccard bags
distance proof
Naive Bayes
PageRank matrix
kNN weighted vote
halving proof
```

03:15–04:00 — Light review only

No new content.

Read your A4 sheet twice.
Read proof skeletons once.
Pack your bag.

Then sleep.

Even 4.5–5 hours is better than going into the exam with a destroyed brain.

Tomorrow morning plan

Assuming you wake around **09:00** and leave/go around **11:00**:

09:00–09:20 — Wake + food

No guilt. No panic-analysis. Eat.

09:20–10:00 — Formula recall

Write from memory:

```
TF-IDF
Jaccard
Naive Bayes
precision/recall/F
PageRank
softmax/cross entropy
k-means objective
```

10:00–10:35 — Proof recall

Write only skeletons:

```
L∞ distance
edit distance
Jaccard distance
halving mistakes ≤ log2(n)
F ≤ (P+R)/2
sum deg(v)=2|E|
```

10:35–10:50 — Exam strategy

In the exam:

```
Read all 4 questions first.
Choose the best 3, not the first 3.
Start with the question that has definitions/formulas you know.
For proofs: list axioms first.
For calculations: write setup even if arithmetic is unfinished.
Do not spend more than 35–38 minutes on one question.
```

10:50–11:00 — Stop

Pack and go.

What you are officially allowed to skip now

```
Deep minhashing / LSH
MNIST notebook details
long transcript reading
full slide reading
beautiful rewritten notes
rare SVM / harmonic graph details unless already on your A4
trying to “understand everything”
```

You are not trying to become fluent in the whole module tonight. You are trying to collect marks.

Your minimum viable exam set

Before sleeping, you need these:

```
1. TF-IDF calculation
2. Bonferroni/birthday expectation
3. document-term matrix definition
4. graph adjacency + sum degrees proof
5. Jaccard sets and bags
6. 4 distance axioms
7. L∞ proof
8. edit distance proof
9. Jaccard distance proof
10. kNN weighted/unweighted
11. Naive Bayes
12. precision/recall/F
13. PageRank + spider traps
14. k-means
15. softmax/cross entropy
16. halving proof
```

That is the plan. Start now with **40 minutes max on the rest of Lecture 2**. Do not negotiate with the slides.

Sources

- `MAT700 Mathematical Methods for Data 2023.pdf`, p.1–4.
- `MAT700 Mathematical Methods for Data 2024.pdf`, p.1–4.
- `MAT700 Mathematical Methods for Data 2025.pdf`, p.1–4.
- `Mapping of Lectures with Tutorials and Transcriptions.txt`.
- `3 - MAT700MA4700 Mathematical Methods for Data Mining online class-20260205_151131-Meeting Recording.txt`.

- `4 - MAT700MA4700 Mathematical Methods for Data Mining online class-20260212_151009-Meeting Recording.txt`.
- `5 - MAT700MA4700 Mathematical Methods for Data Mining online class-20260216_151041-Meeting Recording.txt`.
- `6 - MAT700MA4700 Mathematical Methods for Data Mining online class-20260219_151057-Meeting Recording.txt`.
- `7 - MAT700MA4700 Mathematical Methods for Data Mining online class-20260223_151118-Meeting Recording.txt`.

Okay, as you probably can see in the project conversations, I have for each lecture a conversation called study thread. I'm right now on lecture 2 study thread. And I wanna ask each of these study thread AI managers, AI assistants, I mean, whenever I'm trying to study a new subject, give me explanations for only the top parts that matter. For example, in your plan, you said 15, 25 o'clock till 16, 5 o'clock finish lecture 2, only as exam skeleton. Can you give me a pre-made script I can give my assistants to just like look at the slide notes and lecture slides and question banks of it and transcriptions of it and

Show more

Thought for 24s

Yes. The plan is basically right, but I would make **one correction**:

Lecture 2 must include softmax/cross entropy and graph basics, not only model-based learning + TF-IDF. These appear in recent exam papers. The 2025 paper asks TF-IDF, document-term matrix, adjacency matrix, graph degree proof, mathematical framework, and softmax/cross entropy. MAT700 Mathematical Methods for…

The mapping also confirms L2 = mathematical framework, softmax/cross entropy, TF-IDF, document-term matrix, graph basics, Tutorial 1 P2–P9.

MAT700 lecture mapping

So use these prompts.

Master prompt to paste into every study thread

Copy this first:

```
I am in emergency exam mode for MAT700. I do NOT have time for full lecture coverag

Use ONLY the uploaded MAT700 materials: lecture slides, study notes, question bank,

Your job:
1. Identify the highest-yield exam topics for THIS lecture only.
2. Ignore or briefly mention low-yield/background content.
3. Explain only what I need to answer exam questions.
4. Give me short exam-ready wording, formulas, proof skeletons, and calculation pro
5. After explanation, give me a 10-minute closed-book recall checklist.
6. Do not over-explain. Do not teach the whole lecture.
7. If this is too long, split into Part 1, Part 2, Part 3 and stop after each part.
8. Always map each topic to tutorial questions and past-paper style where relevant.
```

Lecture 2 prompt

Use this in your **Lecture 2 Study Thread** now:

```
Emergency exam mode for Lecture 2.

Look at Lecture 2 slides, Lecture 2 study notes, Lecture 2 question bank, Tutorial

I only need the exam skeleton for Lecture 2. Explain ONLY these high-yield parts:

1. Instance-based vs model-based learning.
2. Mathematical framework for learning:
   - data/training examples
   - unknown probability distribution
   - hypothesis space
   - parameters
   - loss function
   - expected loss
   - empirical mean / law of large numbers idea
   - why training minimises empirical loss.
3. Softmax function and cross entropy:
   - definitions
   - what they are used for in classification
   - exam-ready wording.
4. TF-IDF:
   - TF formula
   - IDF formula with log base 2
   - TF-IDF formula
   - how to compute a TF-IDF matrix.
5. Document-term matrix:
   - definition
   - how to convert it to vector representation.
```

```
6. Graph basics:
   - adjacency matrix
   - weighted adjacency matrix if relevant
   - proof skeleton for sum of degrees = 2|E|
   - idea that powers of adjacency matrix count walks, if in this lecture/question

Format:
A. Exam-ready explanation
B. Formula/proof box
C. One worked mini-example if essential
D. Common mistakes
E. 10-minute closed-book recall checklist

Skip all non-exam background unless it directly helps one of these items.
```

Lecture 3 prompt

```
Emergency exam mode for Lecture 3.

Use Lecture 3 slides, Lecture 3 study notes, Lecture 3 question bank, Tutorial 1 P1

Explain ONLY the highest-yield exam parts:

1. Similar-item / near-duplicate problem:
   - what problem is being solved
   - why documents are turned into sets.
2. Jaccard similarity for sets:
   - formula
   - how to compute intersection/union
   - exam-style example.
3. Jaccard similarity for bags:
   - exact formula
   - min-count numerator
   - denominator used in the notes
   - why max/union set formula is wrong for bags.
4. Proof that Jaccard similarity of bags is ≤ 1/2.
5. Shingling:
   - definition
   - how k-shingles are made
   - why it helps detect similar documents.
6. Minhashing:
   - only the exam-relevant intuition and definition
   - probability/minhash connection to Jaccard if required.
7. Jaccard distance only if this lecture's question bank connects it to L3; otherwi

Format:
A. Exam-ready explanation
B. Formula/proof box
```

```
C. Worked example for sets and bags
D. Common mistakes
E. 10-minute closed-book recall checklist

Do not go deep into LSH unless the question bank strongly suggests it.
```

Lecture 3 is correctly mapped to Jaccard sets, Jaccard bags, shingling, minhashing, Tutorial 1 P10–P11, and Tutorial 2 Jaccard P1.

MAT700 lecture mapping

Lecture 4 prompt

```
Emergency exam mode for Lecture 4.

Use Lecture 4 slides, Lecture 4 study notes, Lecture 4 question bank, Tutorial 2 Di

Explain ONLY the highest-yield exam parts:

1. Distance measure definition:
   - four axioms
   - how to check a proposed function.
2. L1, L2, L∞:
   - formulas
   - especially full proof skeleton for L∞ as a distance measure.
3. Edit distance:
   - definition
   - full proof skeleton that it is a distance measure.
4. Jaccard distance:
   - definition d = 1 - similarity
   - proof skeleton that it is a distance measure.
5. Hamming distance:
   - definition and when useful.
6. kNN:
   - algorithm
   - majority vote
   - weighted kNN, especially 1/d or 1/d²
   - tie handling if in notes.
7. Model evaluation:
   - TP, FP, TN, FN
   - precision, recall, F-measure
   - proof that F ≤ (P+R)/2.
8. Naive Bayes:
   - Bayes formula
   - class conditional independence
   - calculation procedure
   - Laplace smoothing only if in notes/question bank.

Format:
```

```
A. Exam-ready explanation
B. Formula/proof box
C. One worked mini-example for kNN and Naive Bayes
D. Common mistakes
E. 10-minute closed-book recall checklist

This is a high-priority lecture. Keep it focused and exam-style.
```

Lecture 4 is one of the most loaded lectures: distance measures, kNN, precision/recall/F-measure, and Naive Bayes are all mapped there.

MAT700 lecture mapping

Lecture 5 prompt

```
Emergency exam mode for Lecture 5.

Use Lecture 5 slides, Lecture 5 study notes, Lecture 5 question bank, Tutorial 3 Li

Explain ONLY the highest-yield exam parts:

1. PageRank purpose:
   - ranking web pages
   - “important pages are linked by important pages”.
2. Directed graph of the web:
   - pages as nodes
   - links as directed edges.
3. Transition matrix M:
   - how to build it
   - column-stochastic meaning
   - common row/column mistake.
4. PageRank formula:
   - v = ((1 - β)e/n) + βMv
   - meaning of β
   - meaning of teleportation.
5. Iterative computation:
   - how to do 1–3 iterations by hand.
6. Spider traps and dead ends:
   - what they are
   - how damping/teleportation fixes them.
7. Any proof-style PageRank question that appears in the question bank/past papers.

Format:
A. Exam-ready explanation
B. Formula box
C. Step-by-step PageRank calculation template
D. Common mistakes
E. 10-minute closed-book recall checklist
```

```
Do not explain the whole history of Google unless it gives a useful exam sentence.
```

Lecture 5 is correctly mapped to PageRank, transition matrix, random surfer, damping factor, dead ends/spider traps, and Tutorial 3 Link Analysis P1–P2.

MAT700 lecture mapping

Lecture 6 prompt

```
Emergency exam mode for Lecture 6.

Use Lecture 6 slides, Lecture 6 study notes, Lecture 6 question bank, first/middle

Explain ONLY the highest-yield exam parts:

1. Clustering:
   - definition
   - unsupervised learning meaning
   - similarity/distance role.
2. Hierarchical clustering:
   - bottom-up/agglomerative idea
   - starting with each point as its own cluster
   - merging closest clusters
   - stopping at k clusters.
3. Cluster distance:
   - centroid distance
   - minimum distance / single-link idea
   - maximum distance / complete-link idea
   - only what the notes require.
4. Euclidean vs non-Euclidean clustering:
   - centroid in Euclidean space
   - medoid/representative point idea for graphs/non-Euclidean data.
5. k-means:
   - algorithm steps
   - input/output
   - objective function
   - convergence idea
   - local minimum problem.
6. Initial centroids:
   - random choice / repeated starts / any method in notes.
7. Exam-ready answer for "Describe k-means and its objective function."

Format:
A. Exam-ready explanation
B. Algorithm box
C. Objective-function box
D. Common mistakes
E. 10-minute closed-book recall checklist
```

```
Skip long examples unless they directly support k-means or hierarchical clustering
```

Lecture 6 is mapped to clustering, hierarchical clustering, centroids/medoids, k-means, objective function, convergence, and initial centroids.

MAT700 lecture mapping

Lecture 7 prompt

```
Emergency exam mode for Lecture 7.

Use Lecture 7 slides, Lecture 7 study notes, Lecture 7 question bank, later part of

Explain ONLY the highest-yield exam parts:

1. Meta-algorithm:
   - definition
   - exam-ready wording.
2. Online learning / experts setting:
   - what experts are
   - prediction over rounds.
3. Halving algorithm:
   - algorithm steps
   - assumption that one expert is consistent
   - proof that mistakes ≤ log2(n).
4. Weighted majority algorithm:
   - main idea
   - how weights are updated
   - why it is softer than halving.
5. AdaBoost:
   - only the high-level idea
   - weak learners
   - reweighting mistakes
   - combining weak classifiers.
6. Any formula/proof in the question bank.

Format:
A. Exam-ready explanation
B. Algorithm boxes
C. Halving proof skeleton
D. Common mistakes
E. 10-minute closed-book recall checklist

Prioritise halving and weighted majority over deep AdaBoost unless the question bar
```

Lecture 7 is correctly mapped to meta-algorithms, AdaBoost, online learning, experts, halving algorithm, and weighted majority.

Ultra-short version for any thread

When you are too tired, paste this:

```
Emergency exam mode. For this lecture, use the lecture slides, study notes, questic
```

Double-check of your emergency plan

The plan is not random. It matches the high-frequency exam pattern:

```
Lecture 1:
Bonferroni + birthday expectation.

Lecture 2:
Mathematical framework, softmax/cross entropy, TF-IDF, document-term matrix, graph

Lecture 3:
Jaccard sets/bags, shingling, minhashing.

Lecture 4:
Distance proofs, edit distance, Jaccard distance, kNN, precision/recall/F, Naive Ba

Lecture 5:
PageRank calculation, transition matrix, spider traps.

Lecture 6:
k-means, objective function, clustering.

Lecture 7:
Meta-algorithms, halving, weighted majority, AdaBoost idea.
```

The only things I would **not** spend serious time on tonight are: deep LSH, long Google history, MNIST notebook details, deep SVM/harmonic graph material unless your question bank has it as a ready answer, and long philosophical background.

Sources:

- Mapping of Lectures with Tutorials and Transcriptions.txt
- MAT700 Mathematical Methods for Data 2025.pdf, p.2–4

Okay, I learned pretty much everything in lecture 1, lecture 2, and I'm going over lecture 3. Realistically comparing past year exams, how much of the exam have I covered now?

Consider that I will probably be able to finish lecture 4, 5, and not be able to read lecture 6 and 7.

Thought for 1m 18s ›

Answer

Realistically: **right now, with Lecture 1 + Lecture 2 solid and Lecture 3 in progress, you have covered about 35–45% of the exam content.**

But emotionally, this is better than it sounds, because you have covered the material that often forms **Question 1 + early subquestions of Question 2/3**: Bonferroni, birthday expectation, TF-IDF, document-term matrix, graph basics, mathematical framework, softmax/cross entropy, and now Jaccard. These appear very strongly in 2024 and 2025.

MAT700 Mathematical Methods for… MAT700 Mathematical Methods for…

The problem is: **Lecture 4 is massive**. Without Lecture 4, you are not yet safe, because distance proofs, kNN, precision/recall/F-measure, and Naive Bayes appear repeatedly. Lecture 4 is mapped to many past-paper questions across 2023, 2024, and 2025.

MAT700 lecture mapping

Rough coverage by past paper

Scenario	2023	2024	2025	Meaning
L1–L2 only	~22/100 paper pool	~40/100	~35/100	Not enough yet
L1–L3 completed	~22–25/100	~50/100	~35–41/100	Better, but still not safe
L1–L5 completed, skipping L6–L7	~69/100	~83/100	~85/100	Strategically quite safe

Because the exam asks you to **answer 3 out of 4 questions**, not all 4, the important number is not full-paper coverage but whether you can build **three answerable questions**. In 2025, if you cover L1–L5, you could answer almost all of Q1, Q2, and most of Q3/Q4, missing mainly k-means and halving/weighted majority.

MAT700 Mathematical Methods for...

If you finish Lecture 4 and 5 but skip Lecture 6 and 7

That is **not ideal**, but it is a workable emergency strategy.

You would have:

```
L1: Bonferroni, birthday problem
L2: TF-IDF, document-term matrix, graph basics, learning framework, softmax/cross e
L3: Jaccard sets/bags, shingling/minhashing basics
L4: distance proofs, kNN, Naive Bayes, precision/recall/F
L5: PageRank, transition matrix, spider traps
```

That covers the strongest repeated middle of the exam.

What you would miss:

```
L6: k-means / clustering
L7: meta-algorithms, halving, weighted majority, AdaBoost
```

Those are definitely examinable: 2025 asks k-means and initial centroids, plus meta-algorithm/halving/weighted majority; 2024 asks k-means and halving; 2023 asks k-means and halving. MAT700 Mathematical Methods for... MAT700 Mathematical Methods for... MAT700 Mathematical Methods for...

But you can still survive if you do a **micro-version** of L6 and L7 later.

What I recommend now

Finish Lecture 3 quickly, then **move to Lecture 4 immediately**.

Do not spend more than:

```
Lecture 3 remaining: 30–40 min
Lecture 4: 90–120 min
Lecture 5: 60–75 min
```

```
Lecture 6 emergency skim: 20 min
Lecture 7 emergency skim: 20 min
```

For L6/L7, do **not read slides**. Just memorise:

```
K-means:
algorithm steps + objective function + initial centroids

Halving:
algorithm + proof mistakes ≤ log2(n)

Weighted majority:
weights decrease for wrong experts, softer than halving
```

That tiny skim can rescue 5–15 marks if the paper is awkward.

Current honest status

You are **not exam-safe yet**, but you are **not doomed**.

Your real safety threshold is:

```
Finish L4 + L5 = you become strategically viable.
Add 20 min L6 + 20 min L7 = you become much less fragile.
```

So the next priority is not “finish Lecture 3 beautifully.”
It is:

```
Get Lecture 3 exam formulas → move on → conquer Lecture 4.
```

Sources

- `MAT700 Mathematical Methods for Data 2025.pdf`, p.1–4
- `MAT700 Mathematical Methods for Data 2024.pdf`, p.1–4
- `MAT700 Mathematical Methods for Data 2023.pdf`, p.1–4